



# ★ UI Flow — Personal Loan Management System (Bank)

## 1. Landing / Home Screen

**Purpose:** Introduce product & entry points. (*first impression, conversions*)

### UI Elements

- Header: Logo, Login, Register
- Hero: "Apply for Personal Loan" CTA (primary)
- 3 bullets: Fast approval • 100% online • Low interest
- Footer: Contact / Support links

### Primary Actions

- Click **Apply for Personal Loan** → if not logged in, route to Register/Login
- Click **Login / Register**

### API Hooks

- None (public)

### Notes

- Keep CTA prominent to increase conversions.
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## 2. Register (Multi-field Card)

**Purpose:** Create user account and basic KYC. (*identity + contact capture*)

### UI Elements

- Fields: Full Name, Mobile, Email, Password, Aadhaar, PAN
- OTP modal (on mobile)
- Password strength meter
- Terms & Conditions checkbox
- Submit button

### Validations

- Mobile format & uniqueness (API: `POST /api/v1/auth/check-phone` )
- Email format & uniqueness (API: `POST /api/v1/auth/check-email` )
- Aadhaar & PAN basic format validation client-side
- Password strength (min length, uppercase, number)
- Required fields enforced

### Primary Actions

- Submit → show OTP modal → verify OTP → create account (API: `POST /api/v1/auth/register` )

### Notes

- OTP verification prevents fake accounts; capturing PAN/Aadhaar early helps pre-fill later steps.
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## 3. Login

**Purpose:** Authenticate user. (*secure access*)

### UI Elements

- Input: Email or Mobile + Password
- Forgot password link
- Login button
- Option: Login with OTP

### Validations

- Show error messages for wrong creds
- Account lockout after N failed attempts

### Primary Actions

- Login → obtain JWT (API: `POST /api/v1/auth/login`) → redirect to Dashboard

### Notes

- Display last login timestamp for security awareness.
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## 4. User Dashboard (Landing after login)

**Purpose:** Overview & navigation. *(single pane of glass for user)*

### UI Elements

- Top: profile snippet (name, masked PAN/Aadhaar)
- Cards/tiles:
  - Apply New Loan (primary)
  - My Applications (list)
  - Active Loan / Outstanding (if any)
  - Repayments / Upcoming EMI
  - Support / Tickets
- Activity feed / recent messages

### Primary Actions

- Click **Apply New Loan** → Application Stepper

- Click **My Applications** → Application list (status badges)
- Click **Active Loan** → Loan detail / ledger

### API Hooks

- `GET /api/v1/customers/{id}` for profile
- `GET /api/v1/applications?userId=...` for applications summary

### Notes

- If user has active loan, show prominent outstanding & next EMI to reduce late payments.

## 5. Apply — Stepper (Multi-step form)

**Purpose:** Collect all loan application data in digestible steps. (*reduces drop-off*)

### Step A — Personal Info (prefill from profile)

**Fields:** Name (readonly), DOB (auto-calc age), Address, Employment Type (Salaried/Self-employed), Employer name, Office email (optional)

**Validation:** Age must be 21–60 (client + server)

**Action:** Next (API: `PUT /api/v1/customers/{id}` to update profile)

### Step B — Income Details

**Fields:** Monthly gross income, Other income, Existing EMIs total

**Validation:** Numeric checks; FOIR calc preview

**Action:** Next

### Step C — Loan Details

**Fields:** Requested amount, Tenure (months), Purpose (dropdown)

**Validation:** Amount between min/max configured by admin

**Action:** Show EMI estimate (client-side) using EMI formula; Next

## Step D — Eligibility Check (System-run)

**UI:** “Checking your eligibility...” loader (1–2s), then checklist:

- Age: OK / Fail
- Income: OK / Fail
- FOIR: OK / Fail
- CIBIL (mock): OK / Fail
- Existing loan: OK / Fail

**API Hooks:** `POST /api/v1/loan/check-eligibility` returns eligibility + reasons

**Outcome:** If eligible → allow proceed to Documents; If not → show reasons and cool-down info

*(Why: prevents wasted document uploads and improves conversion)*

## Step E — Documents Upload

**UI:** Upload cards for PAN, Aadhaar, Salary slips (3 months), Bank Statement optional, Photo

**Features:** Drag & drop, progress bar, thumbnail preview, re-upload

**Validations:** File types, size ≤ 5MB

**API Hooks:** `POST /api/v1/applications/{id}/documents` multipart

**Action:** Submit application → Status: APPLIED / DOC\_PENDING

## Step F — Review & Submit

**UI:** Read-only summary of all steps with Edit buttons

**Action:** Submit → triggers `POST /api/v1/applications` and backend eligibility + CIBIL fetch

### Notes

- Keep progress saved on each step (localStorage + server autosave).
- Provide inline help (tooltips) for fields like FOIR.

## 6. My Applications (List + Detail)

**Purpose:** Let user track all applications.

### UI Elements

- Table / Cards with Application ID, Amount, Tenure, Status badge, Last updated, CTA: View Details
- Filters: Status, Date range

### Application Detail

- Show stepper-style timeline (APPLIED → ELIGIBLE → DOC\_PENDING → ...)
- Document thumbnails + verification status
- Sanction letter (if any) with Accept/Reject buttons
- Buttons: Raise Query, Withdraw Application (if allowed)

### API Hooks

- `GET /api/v1/applications/{id}`
- `GET /api/v1/applications/{id}/documents`

### Notes

- Display exact dates for each status change for auditability.
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## 7. Notifications & Messages Panel

**Purpose:** Central place for system messages.

### UI Elements

- Inbox-like list: type (SMS/Email/Push), message preview, timestamp
- Read/unread indicator

### Notes

- Important notifications (sanction, disbursements) should trigger modal popups too.
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## 8. Support / Ticketing UI

**Purpose:** Allow customer to create & track issues.

### UI Elements

- Create Ticket form: Subject, Category, Description, Attachments
- Ticket list with status, assigned team, chat thread

### API Hooks

- `POST /api/v1/tickets`
- `GET /api/v1/tickets?userId=...`

### Notes

- Auto-log application ID if ticket created from application page.

## 9. Loan Detail / Ledger Screen (Critical)

**Purpose:** Show amortization ledger and repayment history. *(legal + audit + clarity for user)*

### UI Elements

- Loan summary: Principal, Disbursed date, Interest rate, Outstanding balance
- Next EMI card: amount, due date, pay now button
- Ledger table (paginated): Installment No, Due Date, EMI, Interest, Principal, Paid Date, Status
- Download PDF (ledger & amortization table)
- Prepayment / Foreclosure CTA (if allowed)

### Actions

- Pay EMI → open payment modal (gateway)
- Request prepayment → calculate foreclose amount (API: `POST /api/v1/loans/{id}/prepayment-quote` )

### API Hooks

- `GET /api/v1/loans/{loanId}/ledger`
- `POST /api/v1/emi/{scheduleId}/pay`

### Notes

- Mask sensitive identifiers; show complete audit trail for ledger updates.
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## 10. Payment Modal & Flow

**Purpose:** Facilitate EMI payments.

### UI Elements

- Payment amount (editable if partial allowed), Payment method selector (UPI, Netbanking, Card)
- Redirect to gateway or in-page widget
- Success / Failure screens with transaction reference

### API Hooks

- `POST /api/v1/payments/initiate` → returns gateway token/redirect
- `POST /api/v1/payments/webhook` (backend)

### Notes

- On success: update ledger & send receipt via email/SMS.
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## 11. Loan Officer UI (Back-office)

**Purpose:** Document verification & preliminary review.

### UI Elements

- Inbox / queue: Assigned applications (filters: DOC\_PENDING, RETURNED)
- Application detail view: timeline, documents (large viewer), customer profile, CIBIL, FOIR
- Actions: Approve docs, Return for correction (message template), Reject (reason)



- Quick call button (log → call notes)

### API Hooks

- `GET /api/v1/staff/applications?status=DOC_PENDING`
- `PUT /api/v1/documents/{id}/status`

### Notes

- Provide document comparison tools (original vs re-upload) and tamper indicator.
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## 12. Underwriter UI (Back-office)

**Purpose:** Risk evaluation & final decision.

### UI Elements

- Queue with higher-priority filters (high amount, low CIBIL)
- Risk dashboard: FOIR, inbound enquiries, credit trend
- Decision panel: Approve (customize interest), Reject, Refer to Admin
- Audit comment box for decision rationale

### API Hooks

- `POST /api/v1/applications/{id}/action` (action=UNDERWRITER\_APPROVE/REJECT)

### Notes

- Decisions must be time-stamped and auditable.
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## 13. Admin / Disbursement UI

**Purpose:** Sanction letter, final checks, and disbursement.

### UI Elements

- Sanction generation modal (templates)
- Agreement & e-sign tracking (status)

- Disbursement queue: Ready to disburse (bank validation status)
- Manual override buttons (with reason)

### API Hooks

- `POST /api/v1/admin/sanction/{applicationId}`
- `POST /api/v1/admin/disburse/{loanId}`

### Notes

- Keep strict role-based access; show approvals chain before disbursement.
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## 14. Collection UI (Minimal for MVP)

**Purpose:** Track overdue EMIs and manage reminders.

### UI Elements

- Overdue list with aging buckets
- Customer contact info and ledger view
- Actions: Mark PTP, Log call, Escalate to legal (if > N days)

### API Hooks

- `GET /api/v1/collections/overdue`
- `POST /api/v1/collections/{loanId}/action`

### Notes

- For minimal app, include only manual logging + auto-reminder templates.
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## 15. Common Modals, Components & Patterns

- **Global header** with role-switching (for staff with multiple roles)
- **Universal search** (applications, customers, documents)
- **Toast notifications** for success/errors

- **Confirm modals** for destructive actions (reject, disburse)
- **Audit trail drawer** on every application detail page
- **Accessible forms:** keyboard nav, labels, ARIA

*(Why: consistent UX and accessibility)*

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## 16. Error States & Edge Cases (UI behavior)

- API down: show friendly message + retry option
- File upload failed: allow resume & retry
- Session expired: redirect to login with message
- Race condition (two officers trying to approve): show locking modal or conflict message (optimistic lock)

*(Important to make app robust and reduce frustration)*

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## 17. Acceptance Criteria per Major Screen (short)

- Registration: OTP verified + user created + redirect to Dashboard
- Apply Stepper: data saved per step; eligible check executed; app created
- Documents: upload, preview, and server-side validation pass
- Loan Detail: ledger shows correct amortization for sample loans
- Payment: successful payment updates ledger and sends receipt